

Algorithm of Thoughts Prompting

Instructor

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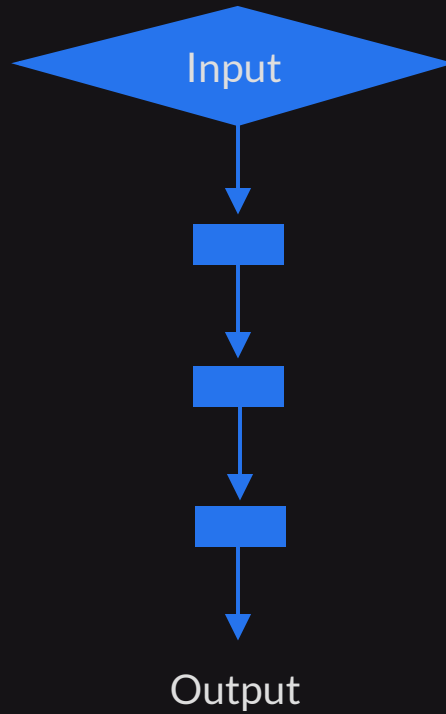
Definition

- An algorithm of Thought (AoT) is a new way to empower large language models with step-by-step problem-solving frameworks.

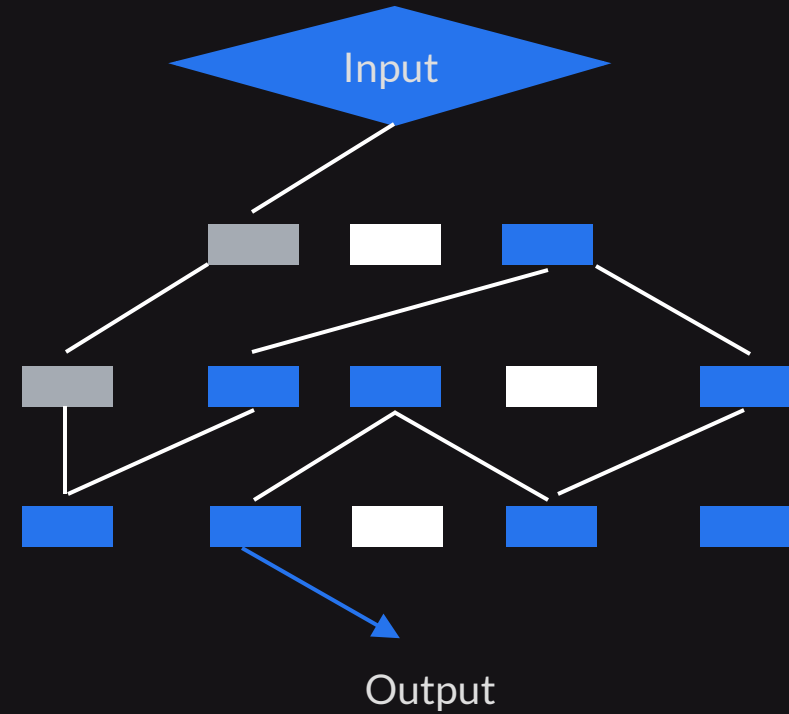
How Does it Work

- **Guided Exploration:** Streamlines the discovery of solutions with minimum queries.
- **Detailed Roadmap:** Offers a comprehensive guide for traversing through problem space.
- **Efficient Problem-solving:** Boasts a sophisticated search strategy that resolves queries more efficiently than CoT.

How Does it Work



Chain-of-Thoughts



Algorithm of Thoughts

Algorithm of Thoughts: Example

Standard Prompting

Model Input

User: 8644
Assistant:
Answer: $(4 + (8 - 6)) * 4 = 24$.
User: 11 10 5 1

Model Output

Assistant:
Answer: $(11 - 1) * (10 - 5) = 24$

Chain-of-Thought

Model Input

User: 8644
Assistant:
8-6 = 2 (left: 4, 4, 2)
4+2 = 6 (left: 6, 4)
6*4 = 24
Answer: $(4 + (8 - 6)) * 4 = 24$.
User: 11 10 5 1

Model Output

Assistant:
11 - 10 = 1 (left: 1, 1, 5)
1 * 1 = 1 (left: 1, 5)
5 * 1 = 5 (left: 5)
Answer: $((11 - 10) * 1) * 5 - 1 = 24$.

Algorithm of Thoughts

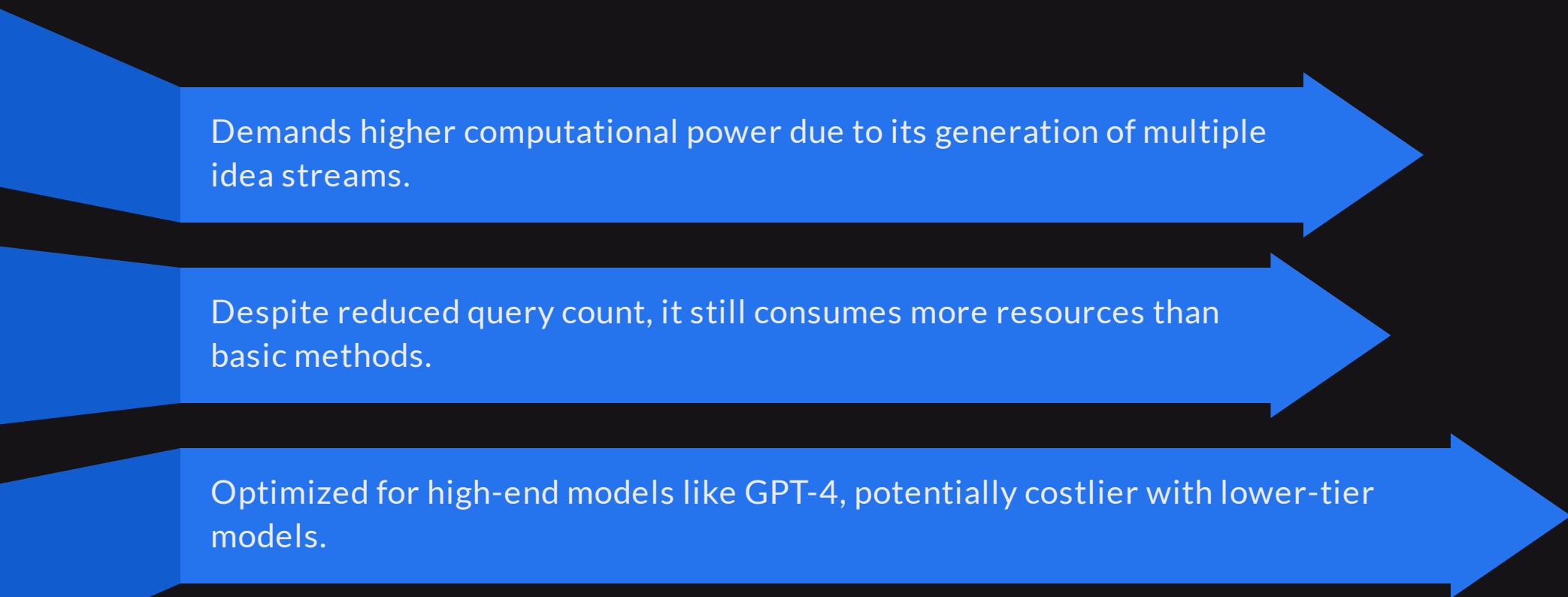
Model Input

User: 8644
Assistant:
1. 4 - 4: (left: 8, 6, 0)
.
.
.
3. 8 - 6: (left: 4, 4, 2)
- 4 + 4: (left: 8, 2) 10, 6, 16, 4
- 4 - 4: (left: 2, 0) 2, 2, 0, undefined
- 4 * 4: (left: 16, 2) 18, 14, 32, 8
- 4 / 4: (left: 2, 1) 3, 1, 2, 2
- 4 + 2: (left: 6, 4) 10, 2, 24 = $6 * 4$ -> found it!
Answer: $(4 + (8 - 6)) * 4 = 24$.
User: 11 10 5 1

Model Output

Assistant:
1. 11-1: (left: 10, 10, 5)
.
.
.
5. 11 = 1: (left: 12, 10, 5)
- 12 + 10: (left: 22, 5) 27, 17, 110, 4.4
- 12 - 10: (left: 5, 2) 7, 3, 10, 2.5
- 12 * 10: (left: 120, 5) 24 = $120 / 5$ -> found it!
Answer: $((11 - 1) * 10) / 5 = 24$.

Cons: Algorithm of Thoughts Prompting



Demands higher computational power due to its generation of multiple idea streams.

Despite reduced query count, it still consumes more resources than basic methods.

Optimized for high-end models like GPT-4, potentially costlier with lower-tier models.

Thank You
